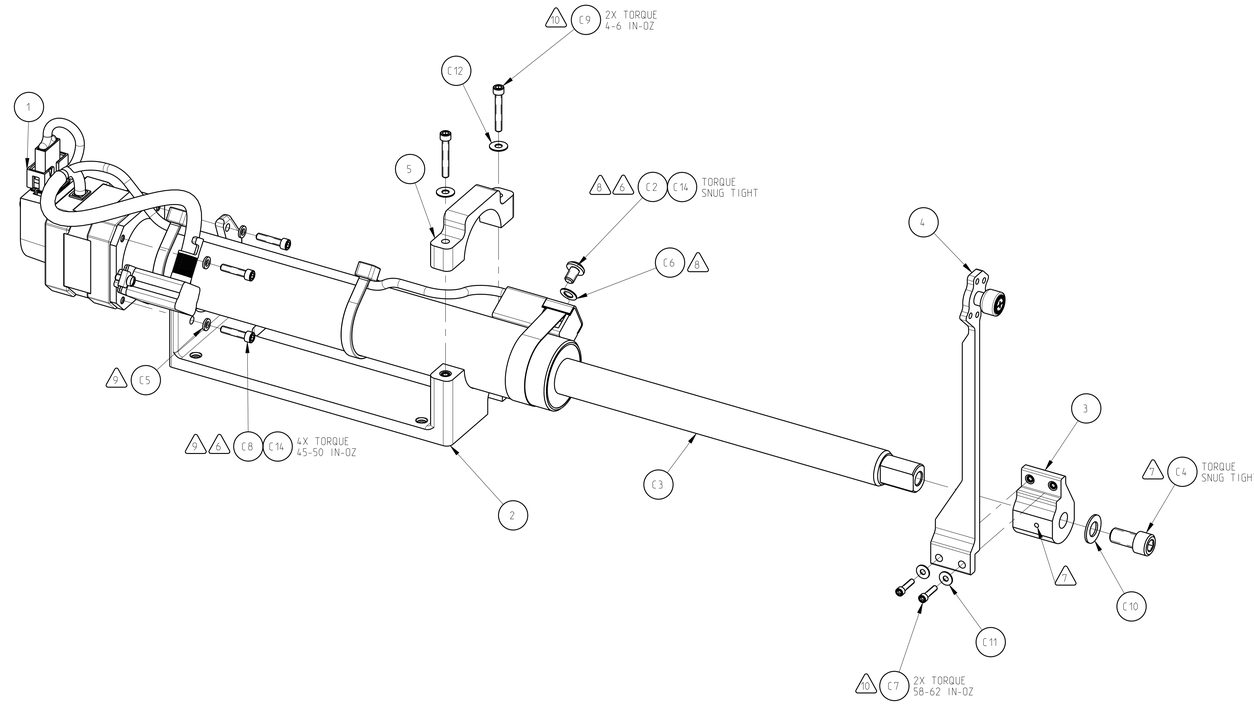
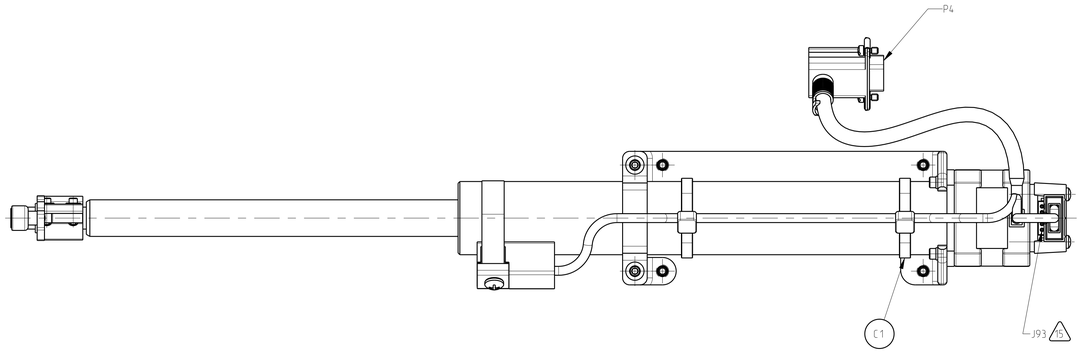
## DRAWING LIST

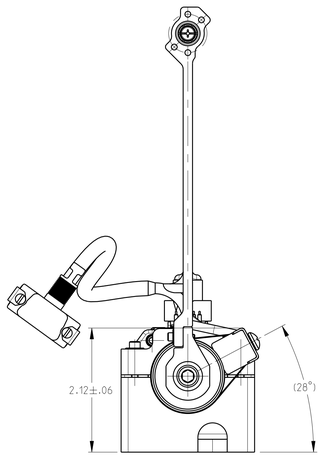
| ITEM | QTY | DRAWING NUMBER | DESCRIPTION |
|---|---|---|---|
| 1 | 1 | S1101EFA1541 | TFP MOTOR TO STEPPER DRIVER HARNESS ASSEMBLY |
| 2 | 1 | S1101MFA1530 | TFP MOTOR BASE ASSEMBLY |
| 3 | 1 | S1101MFA1533 | TFP ARM BASE ASSEMBLY |
| 4 | 1 | S1101MFA1535 | TFP ARM ASSEMBLY |
| 5 | 1 | S1101MFD1532 | TFP MOTOR CLAMP |

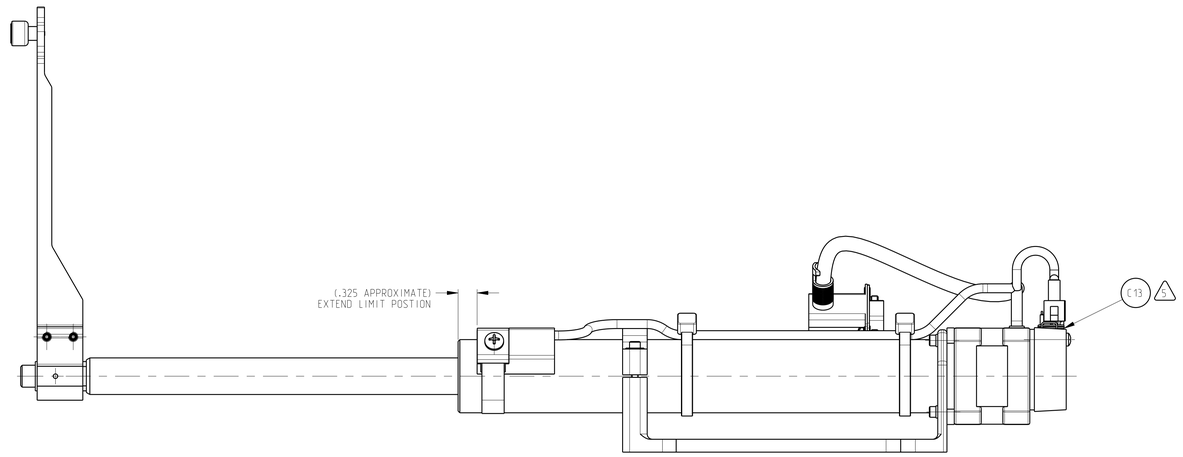
## COMMERCIAL PARTS LIST

| ITEM | QTY | PART NUMBER | DESCRIPTION |
|---|---|---|---|
| C1 | 2 | PLT2S-M702Y | CABLE TIE, HALAR FLUOROPOLYMER, PANDUIT CORP. |
| C2 | 1 | MS51957-41 | SCREW, PAN HEAD, CROSS-RECESSED, 300 SERIES CRES, .164-32 UNC-2A X .250 |
| C3 | 1 | D-A.4AB-HT17068E5-6-IN0 | DIGIT HT17 STEPPER MOTOR LINEAR ACTUATOR, 6 INCH STROKE, ULTRAMOTION |
| C4 | 1 | NAS1351N4-8 | .250-28 UNRF-3A X .500 SOCKET HEAD CAP SCREW |
| C5 | 4 | NAS620C4 | #4 FLAT WASHER REDUCED OUTER DIAMETER |
| C6 | 1 | NAS620C8L | #8 REDUCED THICKNESS FLAT WASHER REDUCED O.D. |
| C7 | 2 | NAS1352N02-6 | #2-56(.086) UNRC-3A X .375 SOCKET HEAD CAP SCREW |
| C8 | 4 | NAS1352N04-8 | #4-40(.112) UNRC-3A X .500 SOCKET HEAD CAP SCREW |
| C9 | 2 | NAS1352N04-12 | #4-40(.112) UNRC-3A X .750 SOCKET HEAD CAP SCREW |
| C10 | 1 | NAS1149C0432R | .250" FLAT WASHER .265" ID X .500" OD X .032" THK |
| C11 | 2 | NAS1149CN216R | NO. 2 FLAT WASHER .099" ID X .250" OD X .016" THK |
| C12 | 2 | NAS1149CN416R | NO. 4 FLAT WASHER .125" ID X .312" OD X .016" THK |
| C13 | A/R | 3145 GRAY | RTV ADHESIVE/SEALANT, GRAY, DOW CORNING |
| C14 | A/R | 222MS | THREADLOCKER, LOCTITE |




NOTES:

1. MATERIAL AND PROCESS CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. ASSEMBLE PER APPROVED PROCESS PLAN FOR INITIAL BUILD.
4. REFER TO S1101EFA1541 FOR WIRING INTERCONNECT.
5. STAKING, BONDING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1.
6. APPLY THREAD LOCKING COMPOUND PER ESTABLISHED PROCEDURES.
7. TFP ARM ALIGNMENT AND STAKING TO BE PERFORMED AT NEXT ASSEMBLY LEVEL. FINAL TORQUE AT NEXT ASSEMBLY. ITEM C4 TO BE THREAD LOCKED AT NEXT ASSEMBLY.
8. REMOVE AND DISCARD EXISTING SCREW AND LOCK WASHER FROM LIMIT POSITION SWITCH AND REPLACE WITH ITEM C2 & C6.
9. REMOVE AND DISCARD EXISTING SCREWS AND FLAT WASHERS FROM STEPPER MOTOR AND REPLACE WITH ITEM C8 & C5.
10. TORQUE VALUES NOTED ON THE DRAWING ARE EFFECTIVE TORQUE ONLY. RUNNING TORQUE IS REQUIRED FOR SELF LOCKING DEVICES AND BOLT FRICTION. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE. RECORD RUNNING TORQUE AND FINAL TORQUE ON DRAWING OR ASSEMBLY PROCEDURE.
11. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE 1, CLASS 3.
12. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
13. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
14. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW. BY TAGGING PER SAE-AS478-35D.
15. REMOVE AND DISCARD HARNESS SUPPLIED WITH ITEM C3 PLUGGED INTO MOTOR ENCODER REFERENCE DESIGNATOR LOCATION P93/J93.

| REV | DESCRIPTION | APPROVED |
|---|---|---|
| A | INCORPORATED ACME-ECO-432. | J.OSRIN 04/27/15<br>A.DRAKE 04/30/15<br>G.FEDEVICH 04/30/15 |

REVISION HISTORY

S1101MFA0100
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INCISE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
.X ± .03
.XX ± .010
.XXX ± .005
5. ANGULAR TOLERANCE ± .5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL. DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E.McCLOSKEY 04/30/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

| | | | |
|---|---|---|---|
| DESIGNER | G.FEDEVICH 02/11/13 | QUALITY | D.SEELEY 02/12/13 |
| REVIEWER | L.FLOOR 02/12/13 | MATERIALS | W.WILKINSON 02/11/13 |
| MECHANICAL | A.DRAKE 02/11/13 | PROJ LEAD | B.BORUNDA 02/28/13 |
| ELECTRICAL | T.COBELI 02/19/13 | | |
| STRUCTURAL | C.WERNER 02/21/13 | | |
| SAFETY | M.BORRELLI 02/11/13 | | |

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

Z/N TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
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CONTRACT NUMBER: NNC09BA02B
PROJECT CODE: SPACE110

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE: TFP ASSEMBLY

| SIZE | CAGE CODE | DWG NO | REV |
|---|---|---|---|
| E | 0SG68 | S1101MFA1529 | A |

SCALE 1/1 | INITIAL REL DATE 02/28/2013 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL: S1101MFA1529

PRO/E VERSION: Wildfire 5.0

DWG NO S1101MFA1529 REV A SH 1

Z1N_E_2012.frm